7	**(a)**		$^{226}_{88}\text{Ra} \rightarrow {}^{222}_{86}\text{Rn} + {}^{4}_{2}\text{He}$	A1
	(b)		$\dfrac{A}{A_0} = \left(\dfrac{1}{2}\right)^{\frac{6.0}{1600\times365}} = 0.99999 \approx 1.0$	B1
			Since the ratio of activity after 6 days to the initial activity is very close to 1, the activity can be considered constant over 6 days. (calculation should be followed with explanation). [Also award 2 marks to reasoned calculation and explanation that the experimental duration of 6 days is insignificant compared to the half-life of 1600 years and so the activity will not decrease significantly.]	B1
	(c)	**(i)**	Alpha particles have very low penetrating power (So the glass wall of cylinder A has to be very thin for the alpha particles to pass through into cylinder B.) No marks are awarded for simple rephrasing of the stem of the question, e.g. “The glass must be thin so that alpha particles can pass through.”	B1
		(ii)	$pV = NkT$ $\Delta N = \dfrac{\Delta p \times V}{kT}$ $\Delta N = \dfrac{0.82\times1.5\times10^{-5}}{1.38\times10^{-23}\times(273+25)}$ $= 2.99\times10^{15}$ Benefit of the doubt was given if the equation used was $N = \dfrac{pV}{kT}$ rather than $\Delta N = \dfrac{\Delta pV}{kT}$	C1 A1
		(iii)	$A \approx \dfrac{N}{t} = \dfrac{2.99\times10^{15}}{6.0\times24\times60\times60}$ $= 5.8\times10^{9}$ Bq	C1 A1

8	(a)	(i)	Polarisation is said to occur when particles in the wave oscillate only in	A1